

Basic Concepts on AUTOSAR Development

Huang Bo, Dong Hui, Wang Dafang and Zhao Guifan

School of Automobile Engineering
Harbin Institute of Technology at Weihai
Weihai, Shandong Province, China
hitdonghui@163.com

Abstract—AUTOSAR(AUTomotiveOpen System ARchitecture) is an open and standardized automotive software architecture, jointly developed by automobile manufacturers, suppliers and tool developers. The AUTOSAR standard will serve as a platform upon which future vehicle applications will be implemented and will also serve to minimize the current barriers between functional domains. To improve cost-efficiency and reusability, AUTOSAR separates application software from the associated hardware. This stage, AUTOSAR is still in the promotion phase, and many countries are committed to its applications, including Japan and South Korea. It will widespread in automotive industry. This paper showed the basic concepts of AUTOSAR and comparisons of some confusing concepts for future in-depth study.

Keywords- AUTOSAR, concept, RTE, VFB.

I. INTRODUCTION

AUTOSAR is a partnership of automotive manufacturers and suppliers working together to develop and establish a de-facto open industry standard for automotive E/E architectures. In AUTOSAR, there are various advanced concepts reflected such as Extensible Markup Language (XML), Application SW-Cs, Virtual Functional Bus (VFB), RTE, BSW modules, and standardized Interfaces. Fig.1 introduces the basic AUTOSAR approach and some basic concepts will be explained.^[1-2]

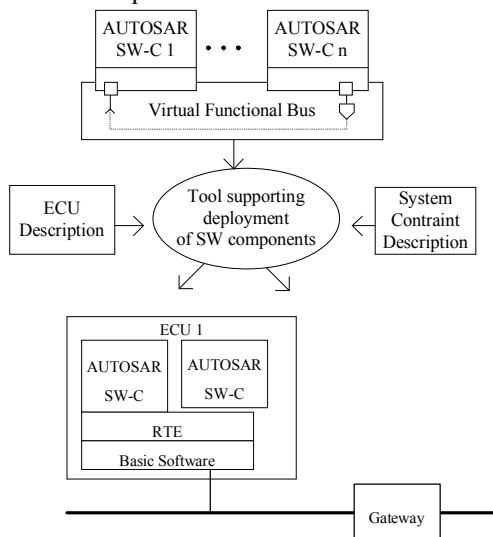


Fig.1: Basic AUTOSAR approach

AUTOSAR development phase can be divided into two categories, such as system development and ECU development.

II. SYSTEM DEVELOPMENT

In system development, each AUTOSAR Software Component encapsulates an application and the Virtual Function Bus allows these components to interact.

A. AUTOSAR SW-C

Each AUTOSAR Software Component (AUTOSAR SW-C) encapsulates part of the functionality of the application which runs on the AUTOSAR infrastructure. Pay attention that application or implementation is separated from infrastructure in the concept of AUTOSAR. Each AUTOSAR SW-C should be assigned to one ECU. The “Sensor/Actuator Software Component” is a special kind of AUTOSAR Software Component which encapsulates the dependencies on specific sensors and/or actuators.^[2-3]

Port : A component has well-defined “ports”, through which the component can interact with other components. Fig.2 shows an example of the definition of a component-type called “SeatHeatingControl”, which controls the heating element in a seat based on several information sources.^[4]

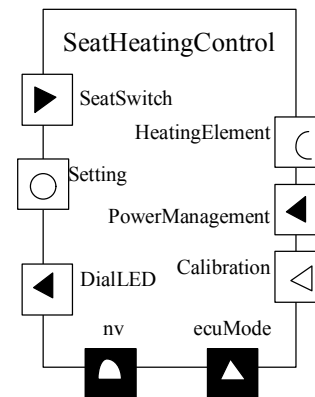


Fig.2: Example of the definition of the component-type “SeatHeatingControl” with eight ports

Port vs Interface: In AUTOSAR, the port and interface are mentioned constantly, in order to make the concepts more clearly and deeply, the distinction is made between them. A port always belongs to exactly one component and represents a point of interaction between a component and other components. The function of port is to provide or require information, therefore ports are divided into “PPort”

and “RPort”. The AUTOSAR Interface concept is introduced to define the services or data that are provided on or required by a port of a component. A port of a component is associated with a “port-interface”. The AUTOSAR Interface contains Client-Server Interface and Sender-Receiver Interface. Fig.3 shows the definition of a simple sender-receiver interface called “SeatSwitch” containing a single data-element called “PassengerDetected”.

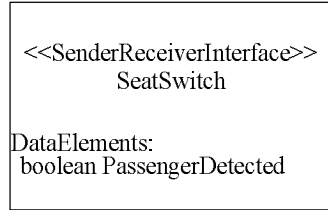


Fig.3: Example of Sender-Receiver Interface “SeatSwitch” with a single data-element

B. Virtual Function Bus (VFB)

The VFB is the abstraction of the AUTOSAR SW-Cs interconnection of the entire vehicle. When the connection between AUTOSAR SW-Cs for a concrete system is defined, the VFB will allow a virtual integration of them in a quite early development phase. The VFB provides a virtual hardware, so AUTOSAR SW-Cs can implement independently from the underlying hardware. Fig.4 shows the representations of Atomic Software Components and AUTOSAR Services connected to the Virtual Function Bus.

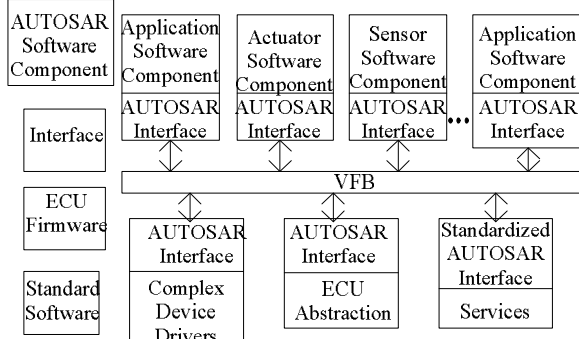


Fig.4: Representations of Atomic Software Components and AUTOSAR Services connected to the Virtual Function Bus.

The VFB provides communication interfaces between SW-Cs and also supports two widely used elementary communication patterns, Client-Server and Sender-Receiver.^[4]

Pay attention that the VFB is an abstraction and is used for plausibility checks virtually in an early time concerning the communication of SW-Cs.

C. The Runnable Concept

The implementation of a component consists of a set of “runnable entities” which is more simply called “runnables”. An atomic SWC can contain one runnable or a large number of runnables. A runnable is a sequence of instructions that can be a very simple piece of code or a complex program. The runnables can be executed by the RTE cyclically or whenever another component invokes an operation on the

port. Fig.5 shows the relationship among the SW-C, RTE and OS.^[5]

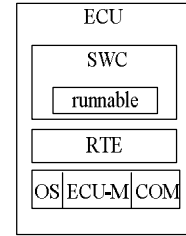


Fig.5: The relationship among the SW-C, RTE and OS.

A task is part of the operating-system(OS) configuration. The OS can decide which task can run on the CPU of the ECU during the run-time. A task provides common resource to runnables such as a context and stack-space. The RTE has the responsibility to make the runnables corresponded to the tasks of the OS. At the same time, the RTE ensure that the runnables are invoked at the correct times.

III. AUTOSAR ECU DEVELOPMENT

AUTOSAR software architecture is composed of three layers, such as application Software components, RTE (Run Time Environment), and Basic Software modules.

A. AUTOSAR Software

The AUTOSAR Software (the layer above AUTOSAR Runtime Environment) consists of AUTOSAR Software Components that are mapped on the ECU.

B. AUTOSAR Runtime Environment

The AUTOSAR Runtime Environment (RTE) is the runtime representation of the Virtual Function Bus for a specific ECU. It has the responsibility to provide a uniform environment to AUTOSAR Software Components. The purpose of this uniform environment is to make the implementation of the software components independent from the communication mechanisms and channels^[4]

VFB vs RTE: VFB and RTE both can separate the application software from the infrastructural software. But they are different concepts, VFB is virtual abstraction, RTE is an entity. The RTE implements the VFB functionality on a specific ECU. In a design step called “Configure System”, the components are mapped on ECUs. Thereby, the virtual connections between the components are mapped onto local connection (within a single ECU) or on network-technology specific communication mechanisms (such as CAN or FlexRay frames). The virtual connections between the components are called VFB. The concrete interface between the components and the rest of the system on an ECU is called RTE.

The RTE provides two communication mechanisms, sender-receiver communication and client-server communication.

In sender-receiver communication, each data-element must be defined to have either “last-is-best” or “queued” semantics. When an interface with data-elements of “last-is-best” semantics. Only 1:n communication (1 sender and n

receivers, with $n \geq 0$) is possible. And the only sender can control the current value of the data-element. For interface with data-elements of “queued” semantics, 1:n communication and n:1 communication are possible. This implies that senders can add values for data-elements to single receiver’s queue.^[5-6]

In client-server communication, the server is a provider of a service and the client is a user of a service. For client-server communication only n:1 ($n \geq 0$, 1 server) communication is supported. When in synchronous communication, clients need to wait for server-side response and stop to invoke arguments during this process. While in the asynchronous communication, clients do not stop running to wait for the results of servers.

C. AUTOSAR Basic Software

Basic Software (BSW) is the standardized software layer, which provides services to the AUTOSAR Software Components and is necessary to run the functional part of the software. It does not fulfill any functional job itself and is situated below the AUTOSAR Runtime Environment. The Basic Software contains standardized and ECU specific components. Fig.6 shows the basic specifics in this layer.

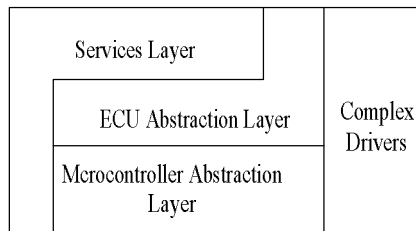


Fig.6: the AUTOSAR basic software.

IV. CONCLUSION

AUTOSAR divides electronic structure into several layers and modules. At the same time of defining the interface between these layers and modules, AUTOSAR also defines software components and standardized hardware platform. To achieve the separation between applications and the base modules which is the objective of AUTOSAR, the ECU internal codes are abstracted into several layers.

This paper showed the basic concepts of AUTOSAR through two directions, such as system development and ECU development. Comparison of several concepts, like port and interface, VFB and RTE, make the definition more clearly.

ACKNOWLEDGMENT

This research is supported by National Science and Technology Major Project (Core electronic devices, high-end general-purpose chip and basic software products 2009ZX01038-002-001). At the same time, the research belongs to the project of Doctorate Fund of the Ministry of Education(20092302120059).

REFERENCES

- [1] www.autosar.org
- [2] Daehyun Kum and Gwang-Min Park, AUTOSAR Migration from Existing Automotive Software, Oct.14-17,2008
- [3] Zhou Zhiguo. The study of diesel engine electronic control module software which is based on AUTOSAR. Zhejiang University.2008
- [4] AUTOSAR Administration. Technical Overview V2.2.1.AUTOSAR GbR.2008
- [5] Vector Informatik GmbH, AUTOSAR_Workshop, Mar.16,2009
- [6] Copyright Elektrobit Corporation 2007 Elektrobit.com, October 2007.